

Evolución

De la antigüedad a nuestros días

Arquitectura de computadoras

Pablo A. Montini
Juan I. Iturriaga
Franco Lanzillotta

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	Burroughs	First machine designed for a high-level language
1964	360	IBM	First product line designed as a family
1964	6600	CDC	First scientific supercomputer
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	Burroughs	First machine designed for a high-level language
1964	360	IBM	First product line designed as a family
1964	6600	CDC	First scientific supercomputer
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Dispositivos Mecánicos

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
			First machine designed for a high-level language
			First product line designed as a family
			First scientific supercomputer
			First mass-market minicomputer (50,000 sold)
			Dominated minicomputers in the 1970s
			First general-purpose 8-bit computer on a chip
			First vector supercomputer
			First 32-bit superminicomputer
			Started the modern personal computer era
			First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Dispositivos Mecánicos

Máquina Diferencial

- solo sumas y restas
- 1 solo algoritmo
- tablas de navegación

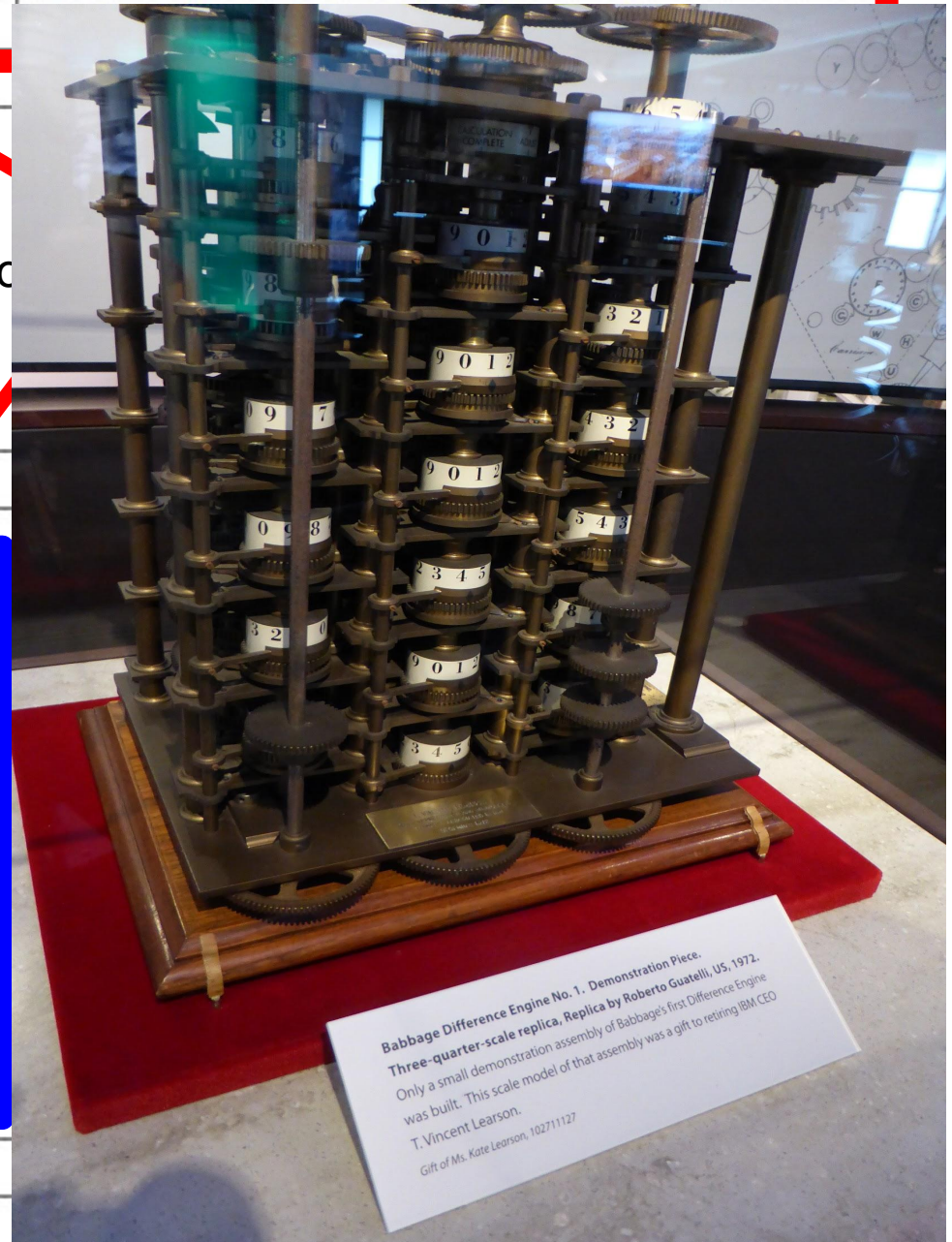
Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	
1943	COLOSSUS	British gov t	
1944	Mark I	Aiken	
1946	ENIAC I		
1949	EDSAC		
1951	Whirlwind I		
1952	IAS	Von	
1960	PDP-1	DEC	
1961	1401	IBM	
1962	7094	IBM	

Dispositivo

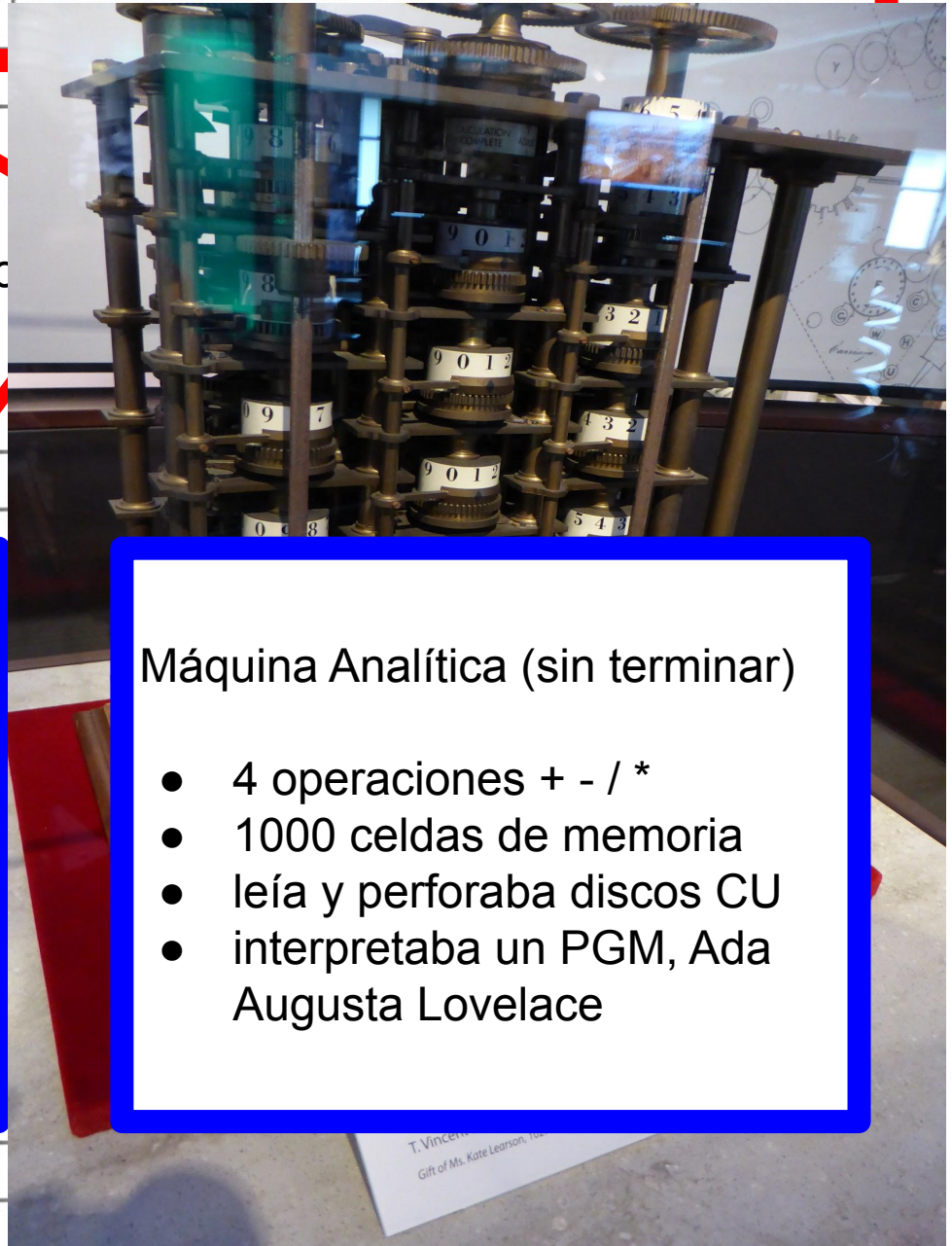
Máquina Diferencial

- solo sumas y restas
- 1 solo algoritmo
- tablas de navegación

1987	SPARC	Sun
1990	RS6000	IBM



Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	
1943	COLOSSUS	British gov't	
1944	Mark I	Aiken	
1946	ENIAC I		
1949	EDSAC		
1951	Whirlwind I		
1952	IAS	Von	
1960	PDP-1	DEC	
1961	1401	IBM	
1962	7090	IBM	



Máquina Diferencial

- solo sumas y restas
- 1 solo algoritmo
- tablas de navegación

Máquina Analítica (sin terminar)

- 4 operaciones + - / *
- 1000 celdas de memoria
- leía y perforaba discos CU
- interpretaba un PGM, Ada Augusta Lovelace

1987	SPARC	Sun
1990	RS6000	IBM

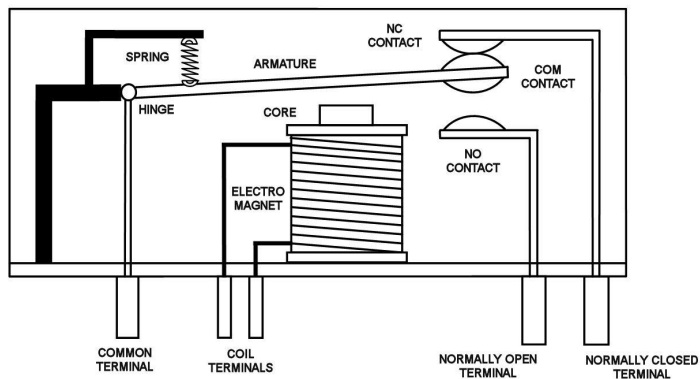
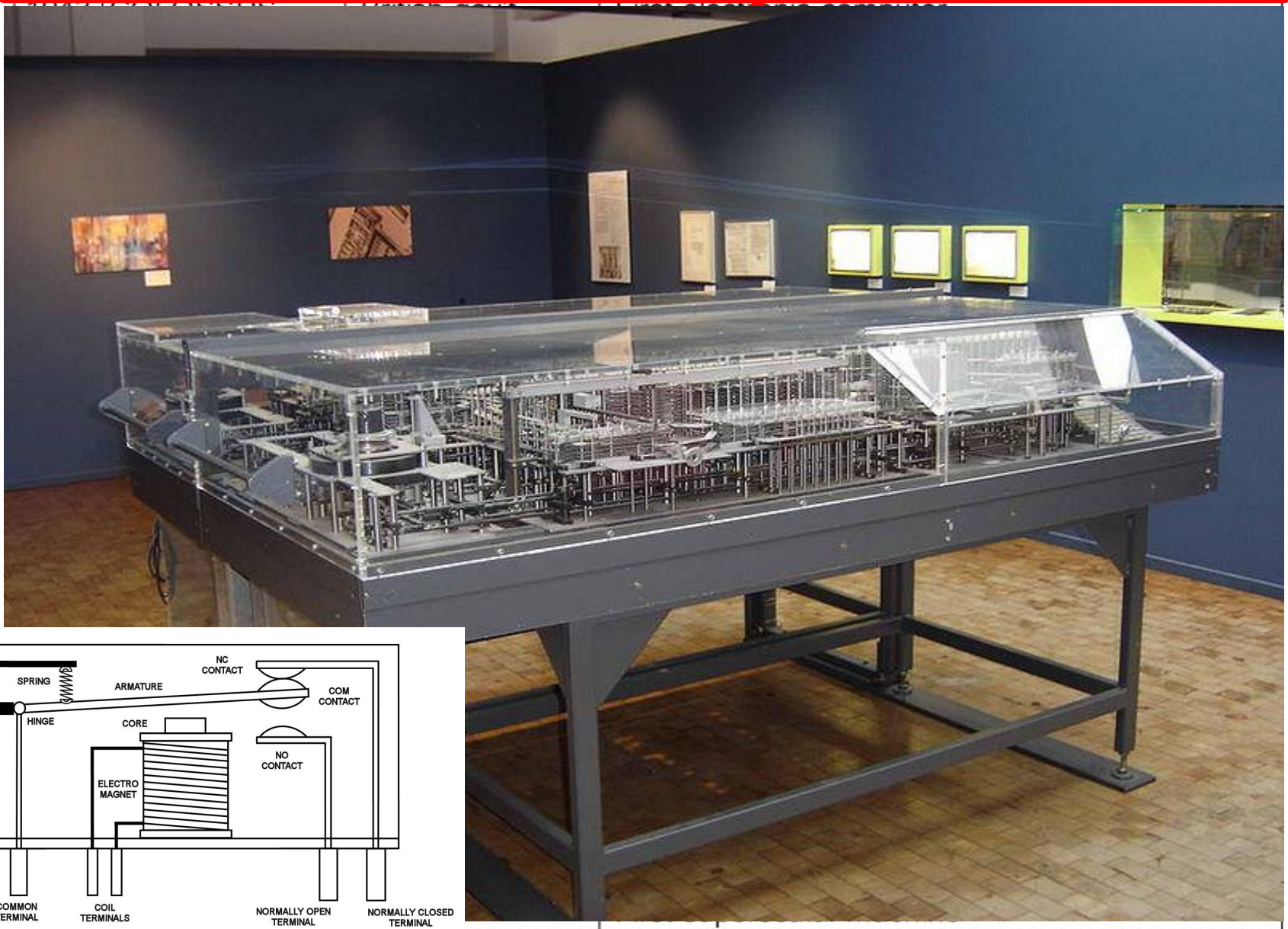
Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	Burroughs	First machine designed for a high-level language
1964	360	IBM	First product line designed as a family
1964	6600	CDC	First scientific supercomputer
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Dispositivos Mecánicos

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine



Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine

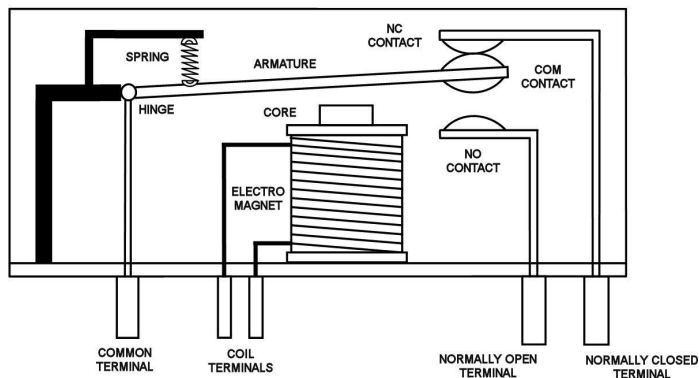


Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine



Z1

- Lógica binaria + FP
- In-out cinta perforada
- 9 instrucciones
- 1 reg acum FP 22 bit
- Memoria 64 palabras FP



Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	Burroughs	First machine designed for a high-level language
1964	360	IBM	First product line designed as a family
1964	6600	CDC	First scientific supercomputer
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Dispositivos Mecánicos

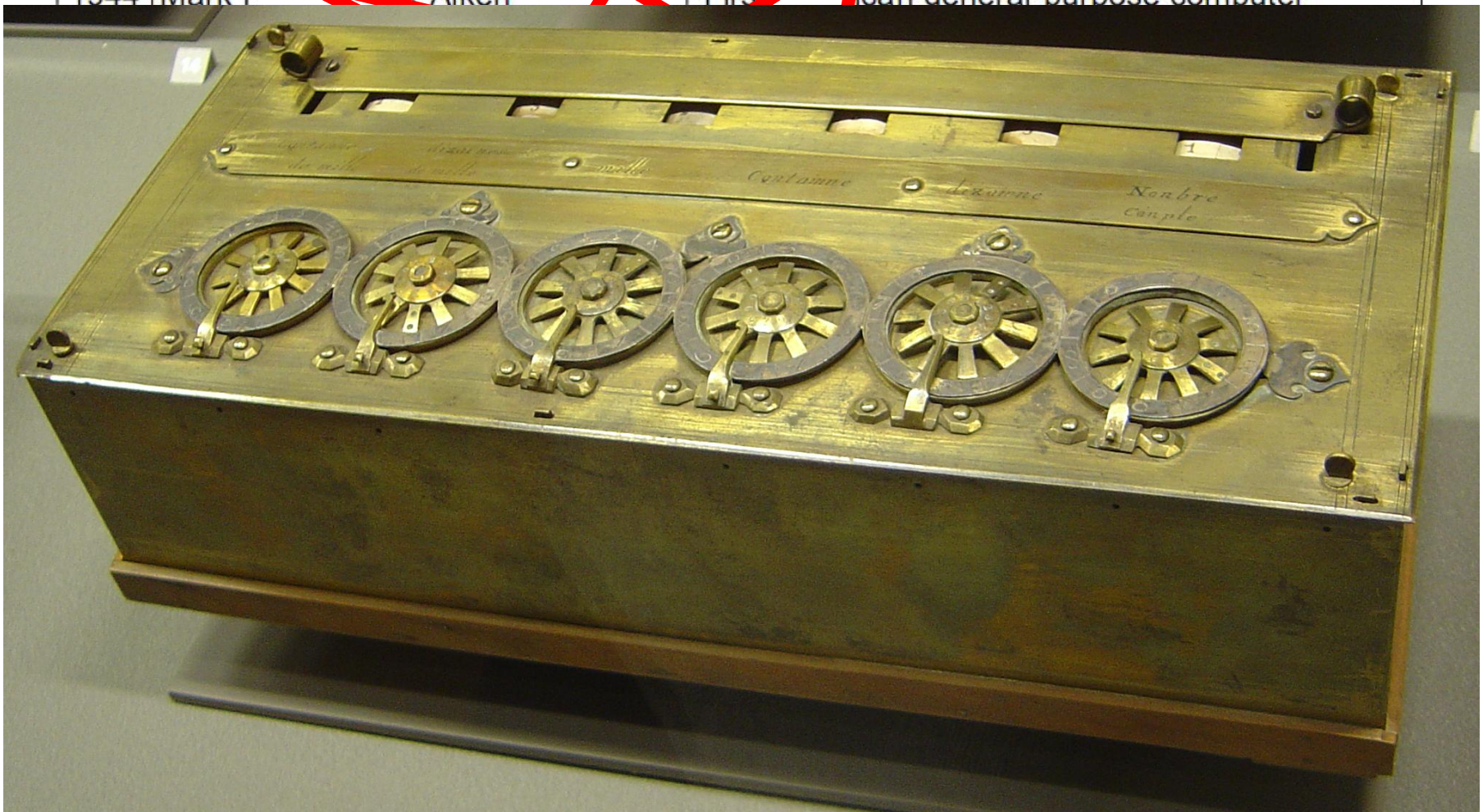
Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s

Dispositivos Mecánicos

1642-1670 Pascalina Pascal sumas y restas ...4 operaciones

1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer



1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s

Dispositivos Mecánicos

1642-1670 Máq. Diferencial Pascal sumas y restas

1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip

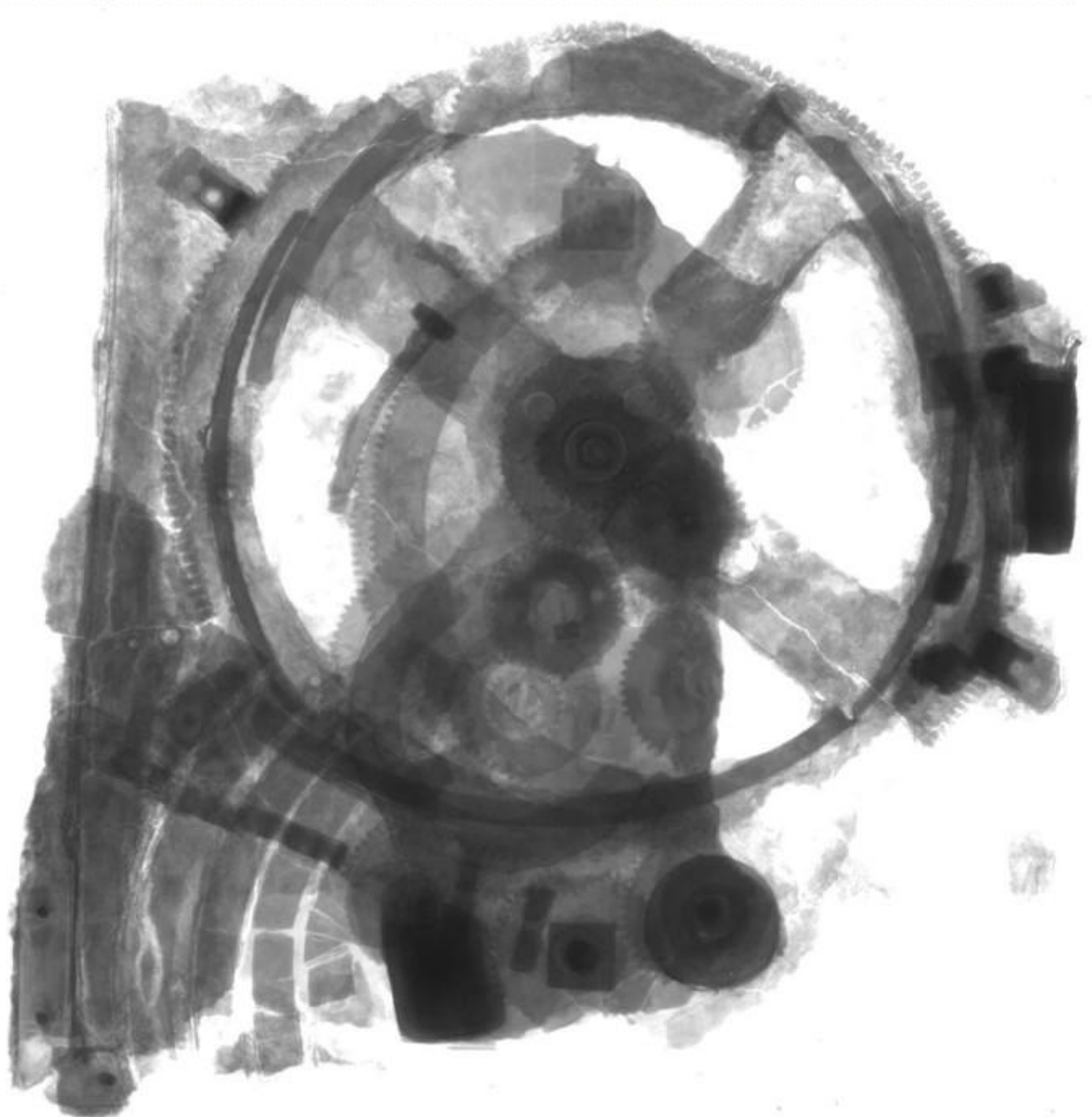
87 AC Antikythera Grecia Engranajes diferenciales, movimiento de 6 planetas, eclipse de sol y luna, fecha de los JJOO

1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

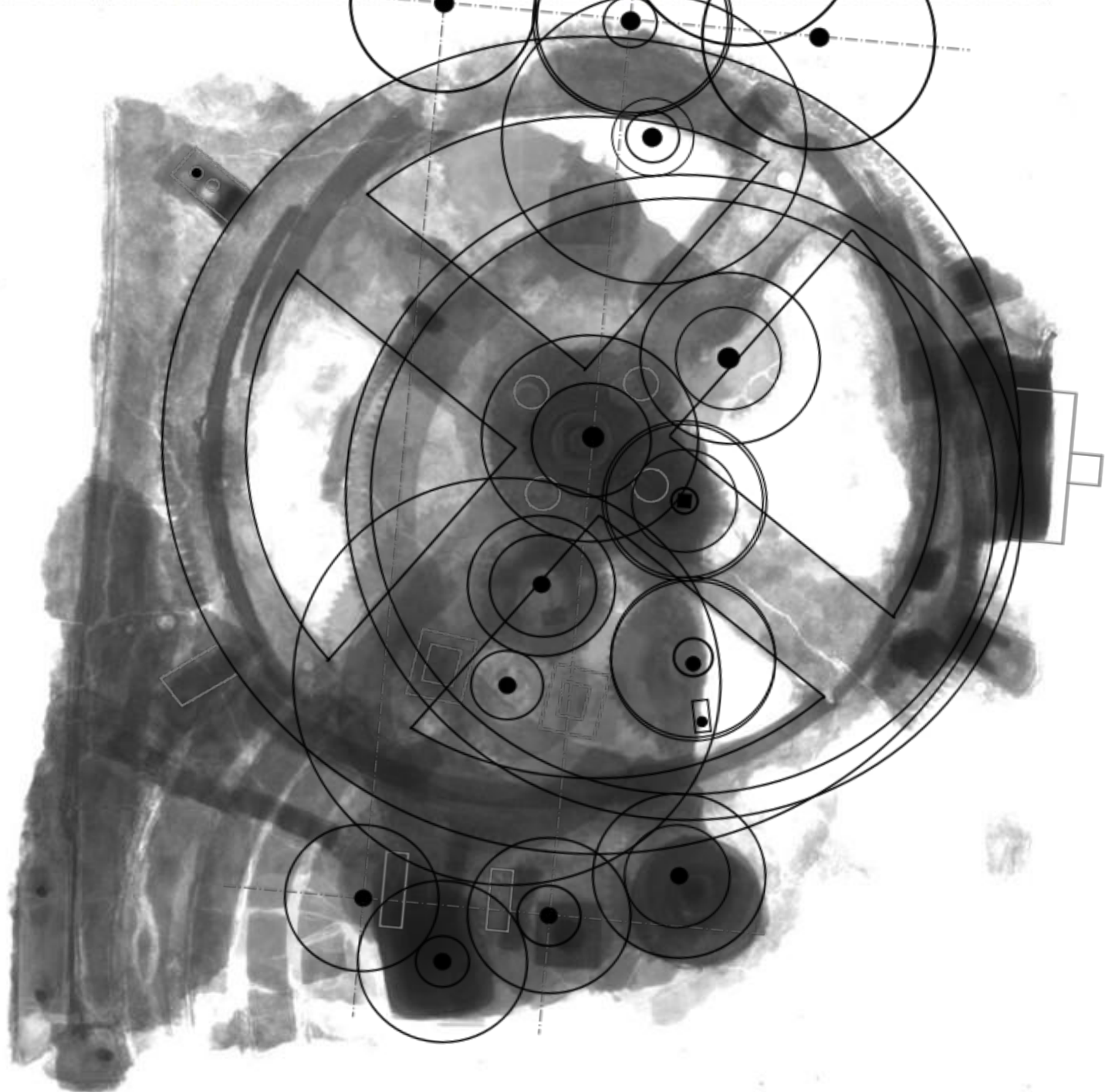
Year	Name
1834	Analytic
1936	Z1
1943	COLOS
1944	Mark I
1946	ENIAC
1949	EDSAC
1951	Whirlwir
1952	IAS
1960	PDP-1
1961	1401
1962	7090
1965	PDP-8
1970	PDP-11
1974	8080
	87 AC
1985	MIPS
1987	SPARC
1990	RS6000



Year	Name
1834	Analytic
1936	Z1
1943	COLOS
1944	Mark I
1946	ENIAC I
1949	EDSAC
1951	Whirlwir
1952	IAS
1960	PDP-1
1961	1401
1962	7094
1965	PDP-8
1970	PDP-11
1974	8080
	87 AC
1985	MIPS
1987	SPARC
1990	RS6000



Year	Name
1834	Analytic
1936	Z1
1943	COLOS
1944	Mark I
1946	ENIAC I
1949	EDSAC
1951	Whirlwir
1952	IAS
1960	PDP-1
1961	1401
1962	7094
1965	PDP-8
1970	PDP-11
1974	8080
87 AC	
1985	MIPS
1987	SPARC
1990	RS6000



Year	Name
1834	Analytic
1936	Z1
1943	COLOS
1944	Mark I
1946	ENIAC
1949	EDSAC
1951	Whirlwir
1952	IAS
1960	PDP-1
1961	1401
1962	7094
1965	PDP-8
1970	PDP-11
1974	8080
	87 AC
1985	MIPS
1987	SPARC
1990	RS6000



Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I		History starts here
1949	EDSAC		Computer
1951	Whirlwind I		
1952	IAS	Von	Lines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s

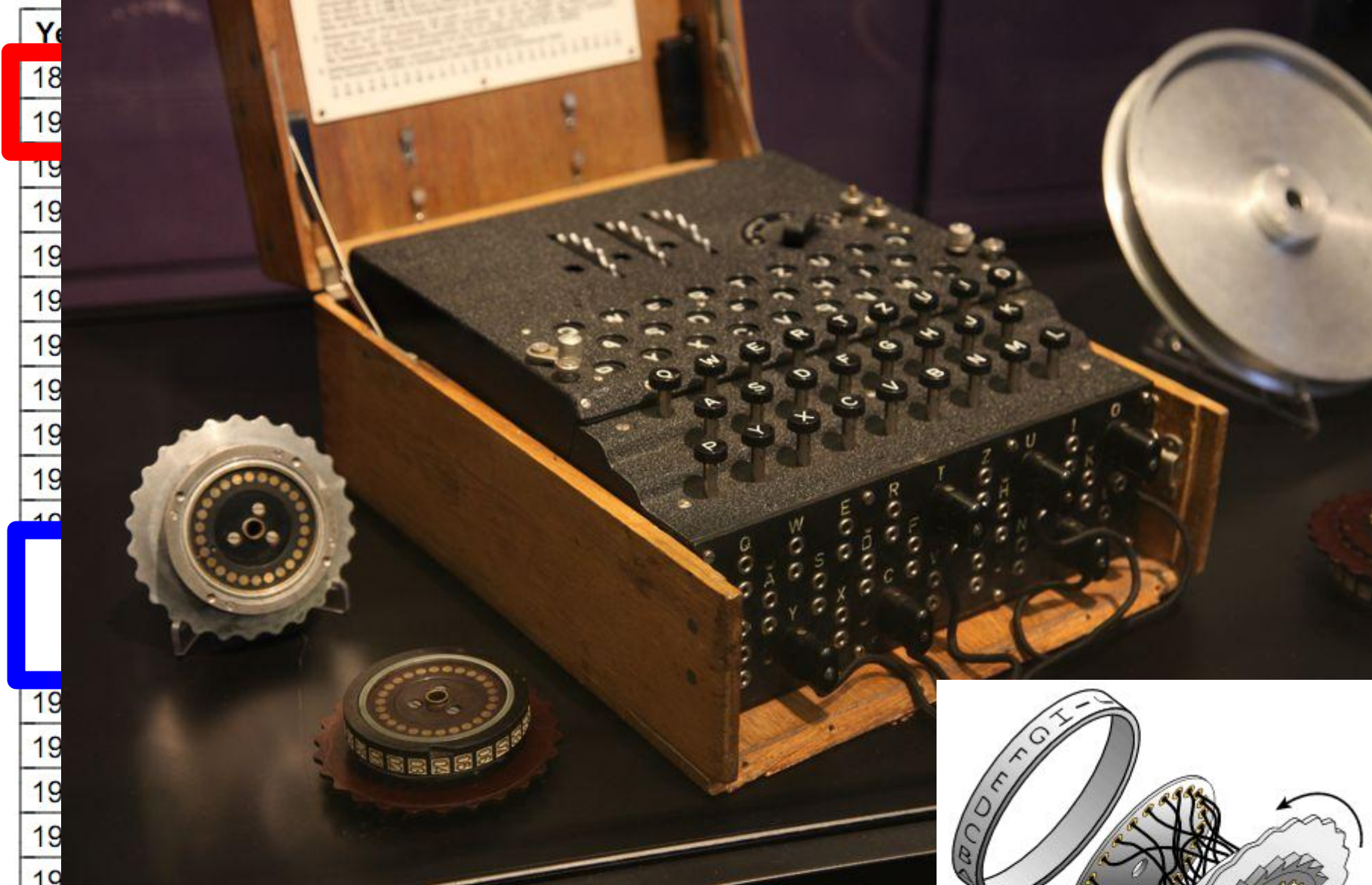
Dispositivos Mecánicos

1940 The Bombe - Alan Turing // Máquina Enigma

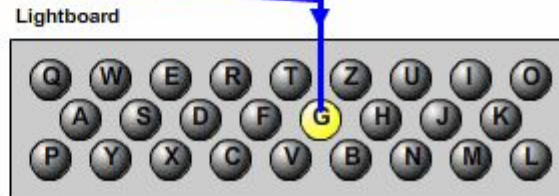
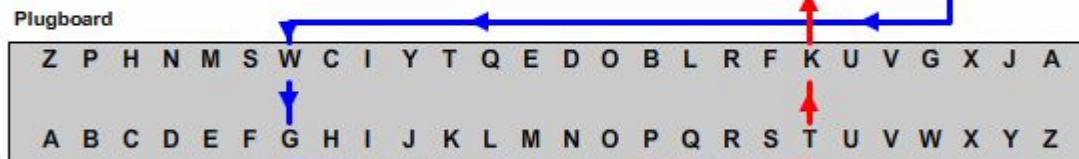
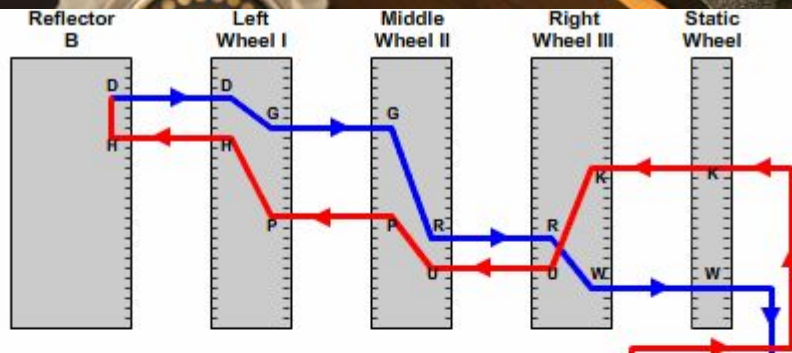
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine



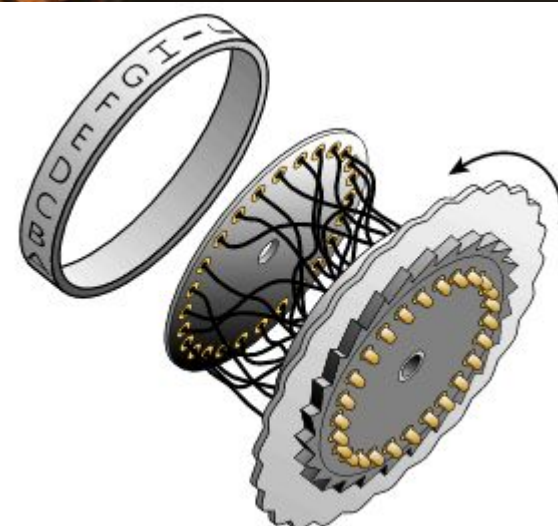
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

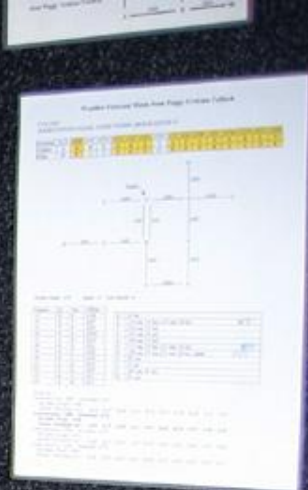
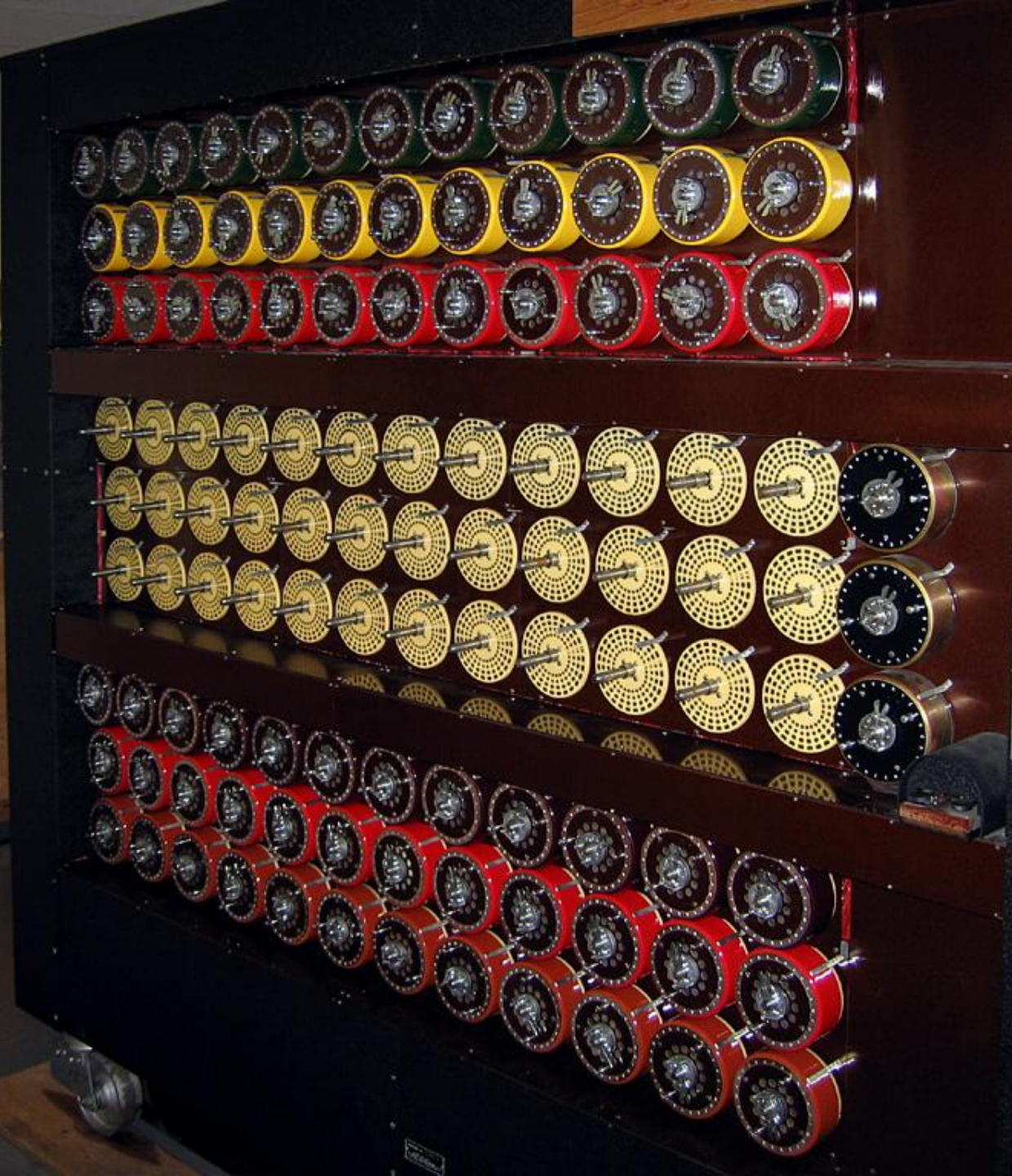


1981	IBM PC	IBM	Started the modern pers
1985	MIPS	MIPS	First commercial RISC n
1987	SPARC	Sun	First SPARC-based RIS
1990	RS6000	IBM	First superscalar machin

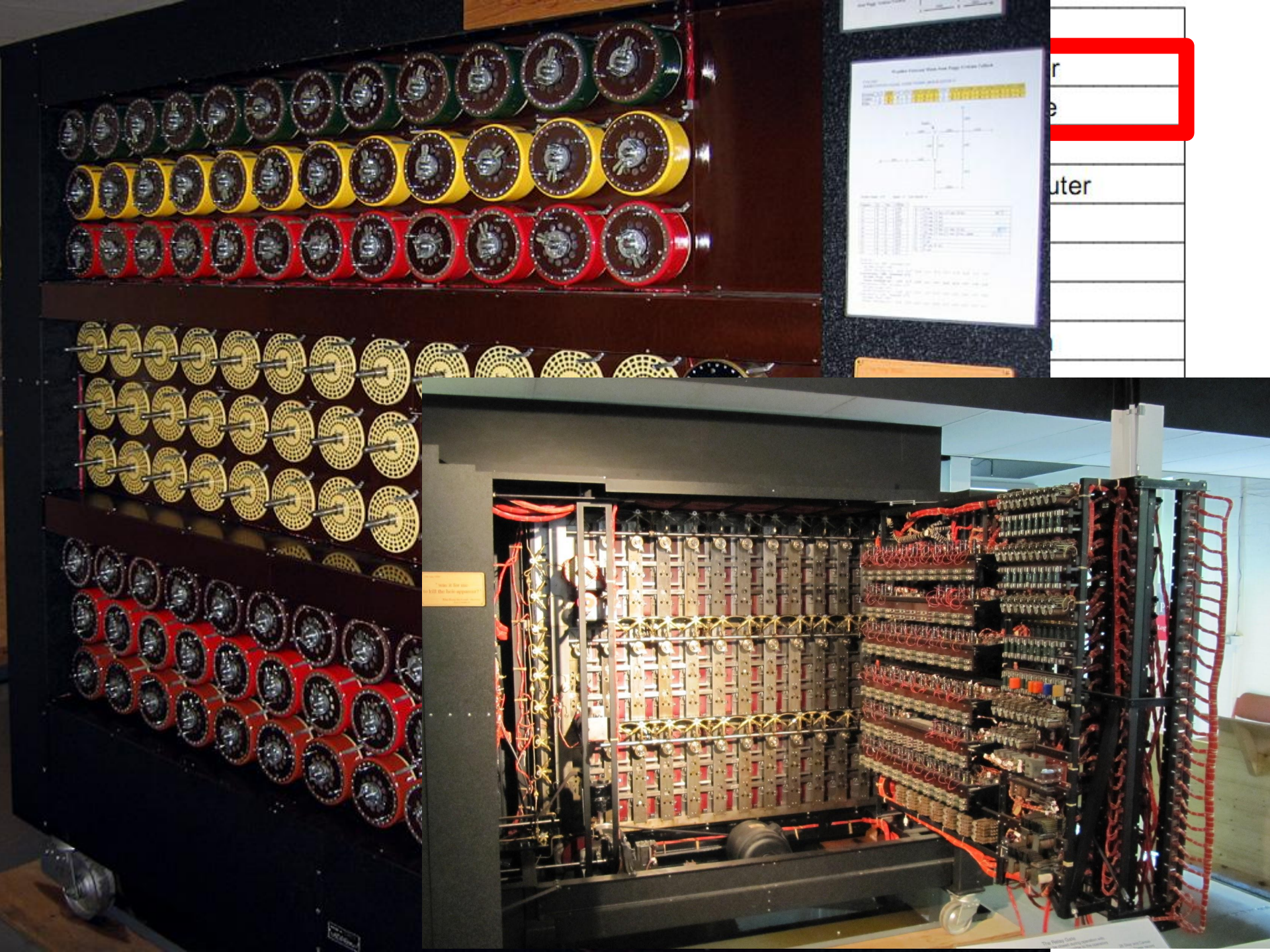


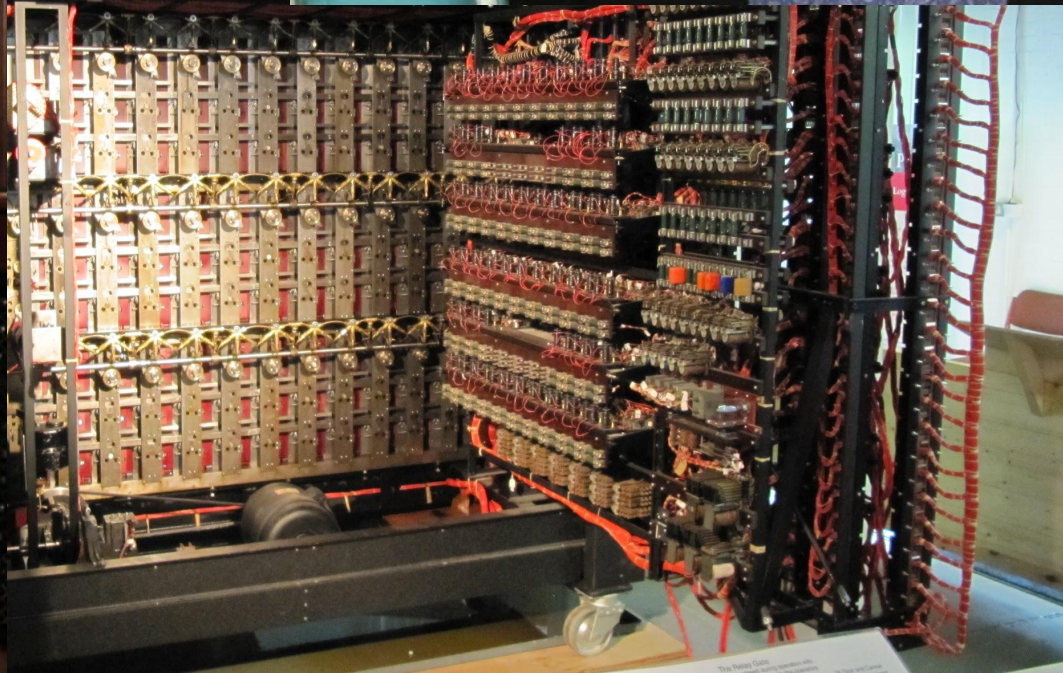
ern pers
RISC n
sed RIS
machin

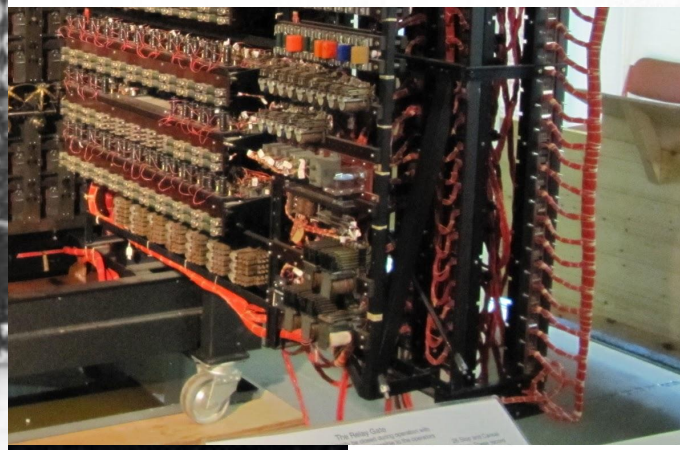




r
e
uter
achine
only 1000
00 sold)
s
n a chip
era

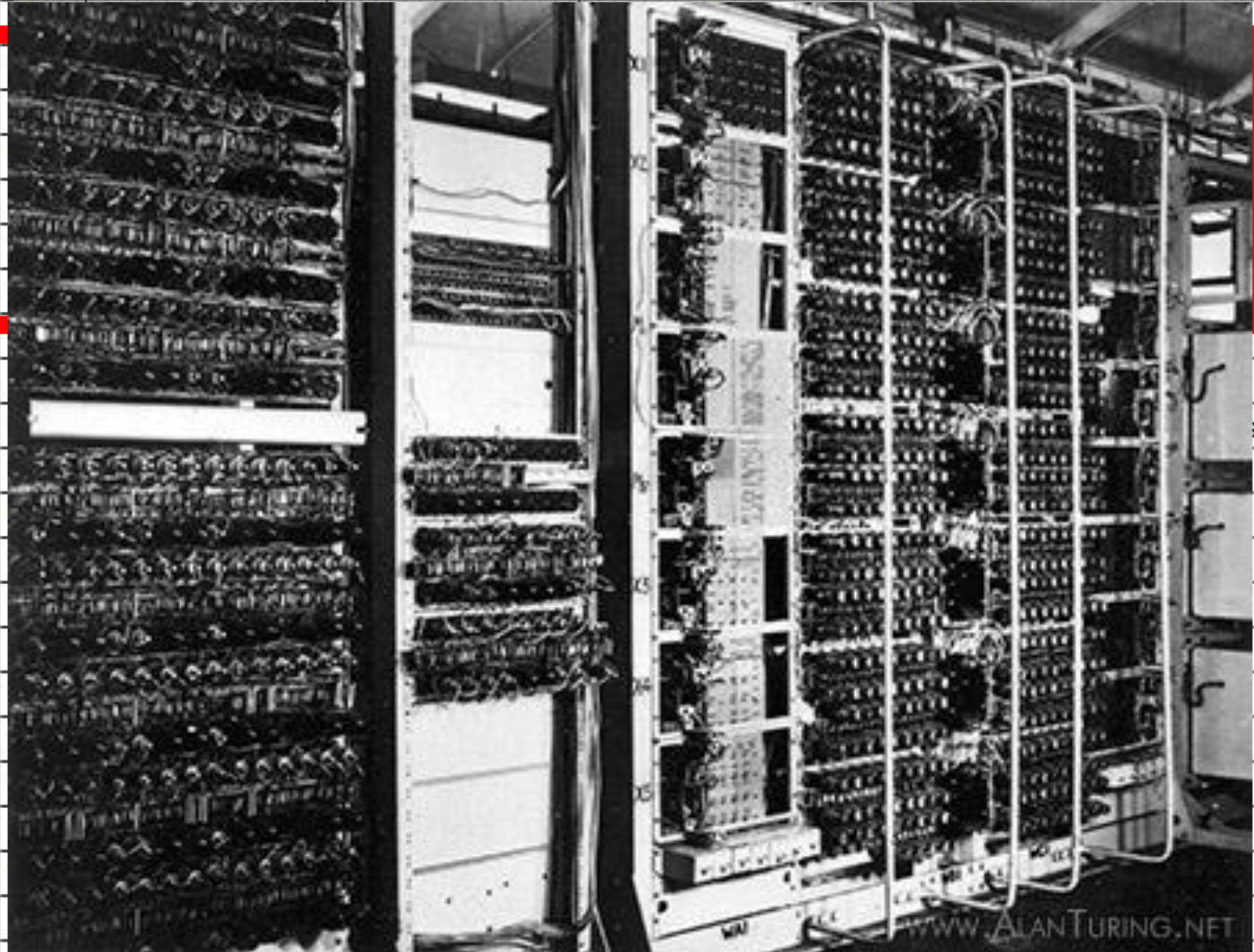




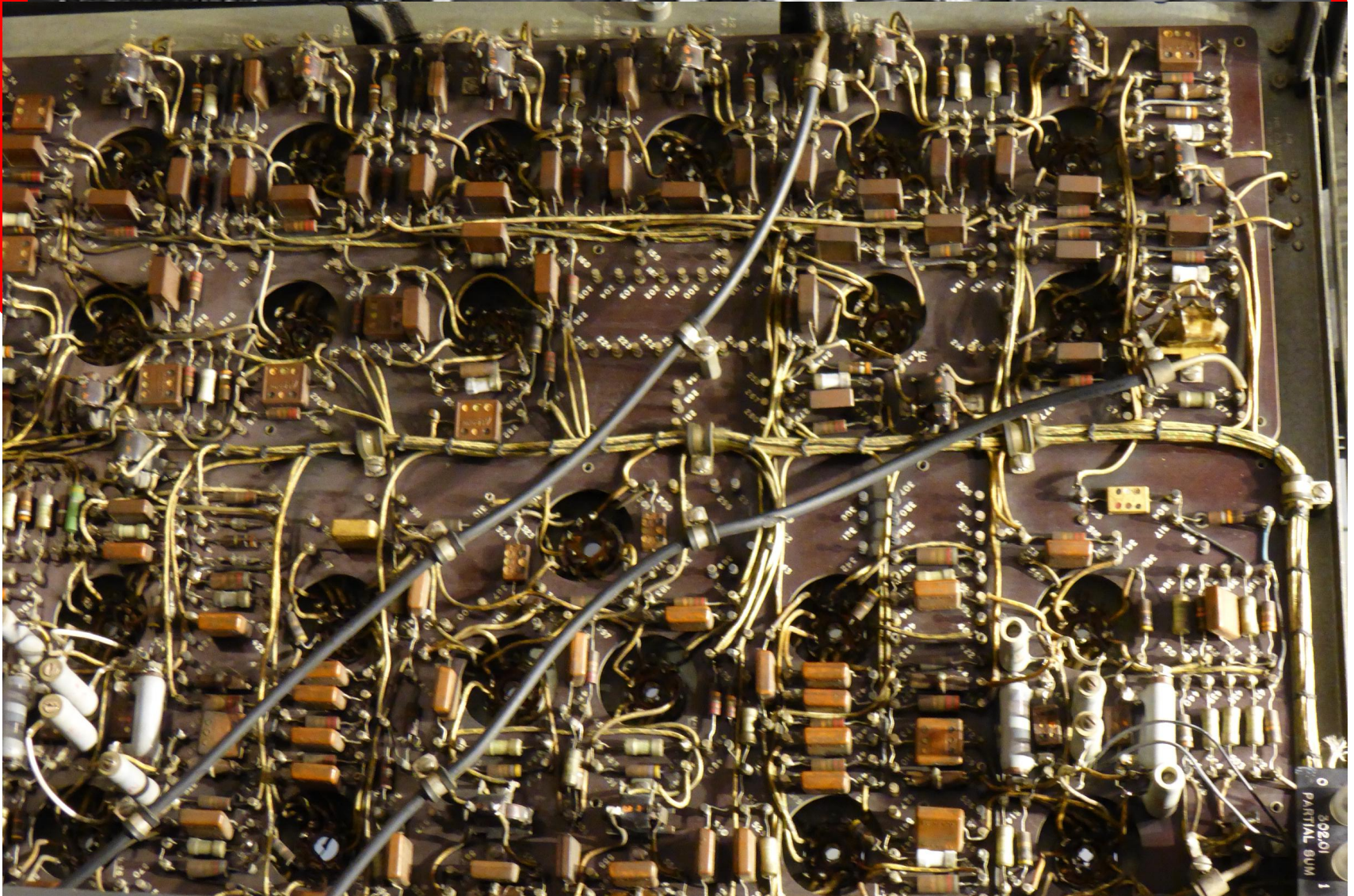


Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	First widely popular small business machine
1962	7094	IBM	First computer for scientific computing in the early 1960s
1963	B5000	IBM	First computer designed for a high-level language
1964	360	IBM	First computer as a family
1964	6600	Cray	First supercomputer
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

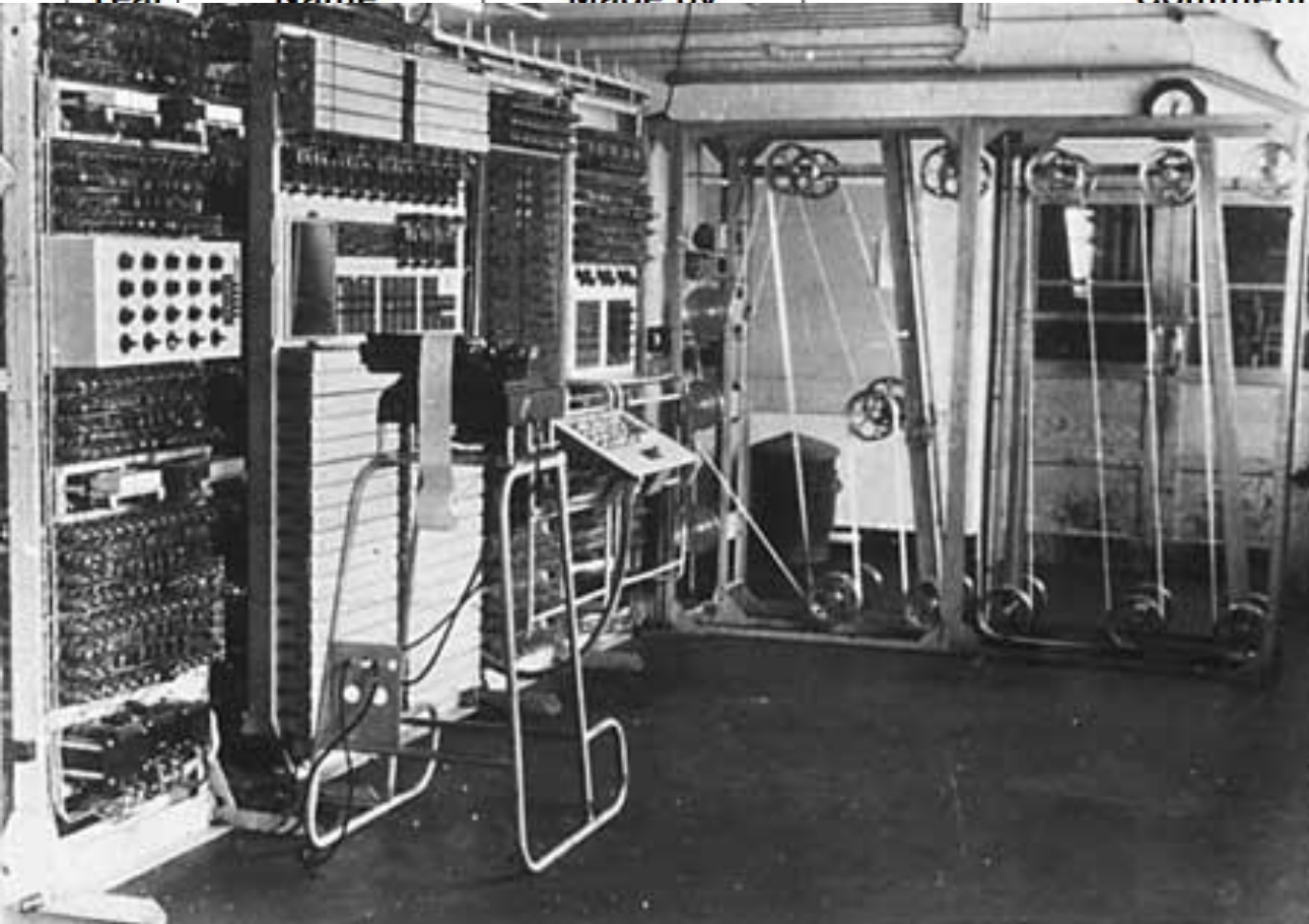
Dispositivos de Válvulas

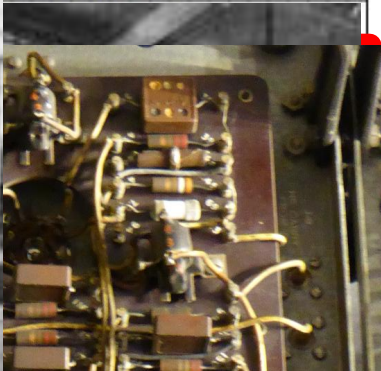
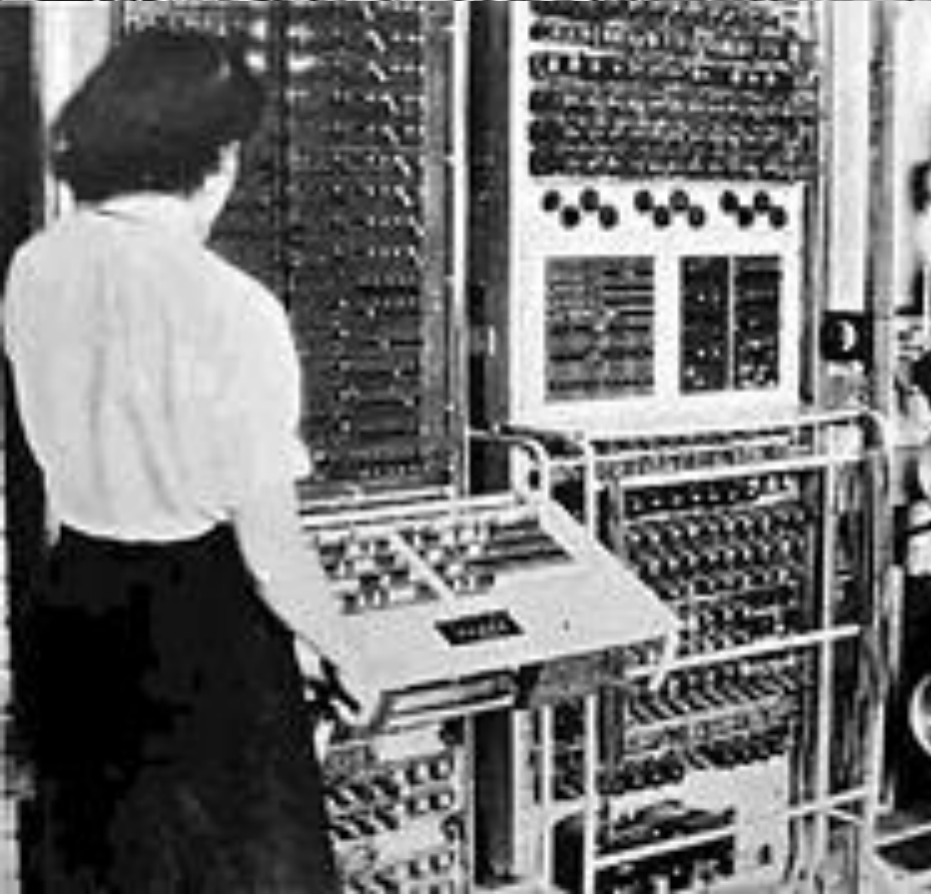
Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
			
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer



1990	RS6000	IBM	First superscalar machine
------	--------	-----	---------------------------

Year	Name	Made by	Comments
			computer
			
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
			computer
			
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	First widely popular small business machine
1962	7094	IBM	First scientific computing in the early 1960s
			First computer designed for a high-level language
			First computer as a family
			First minicomputer
			First mass-market minicomputer (50,000 sold)
			Dominated minicomputers in the 1970s
			First general-purpose 8-bit computer on a chip
			First vector supercomputer
			First 32-bit superminicomputer
			Started the modern personal computer era
			First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

ENIAC

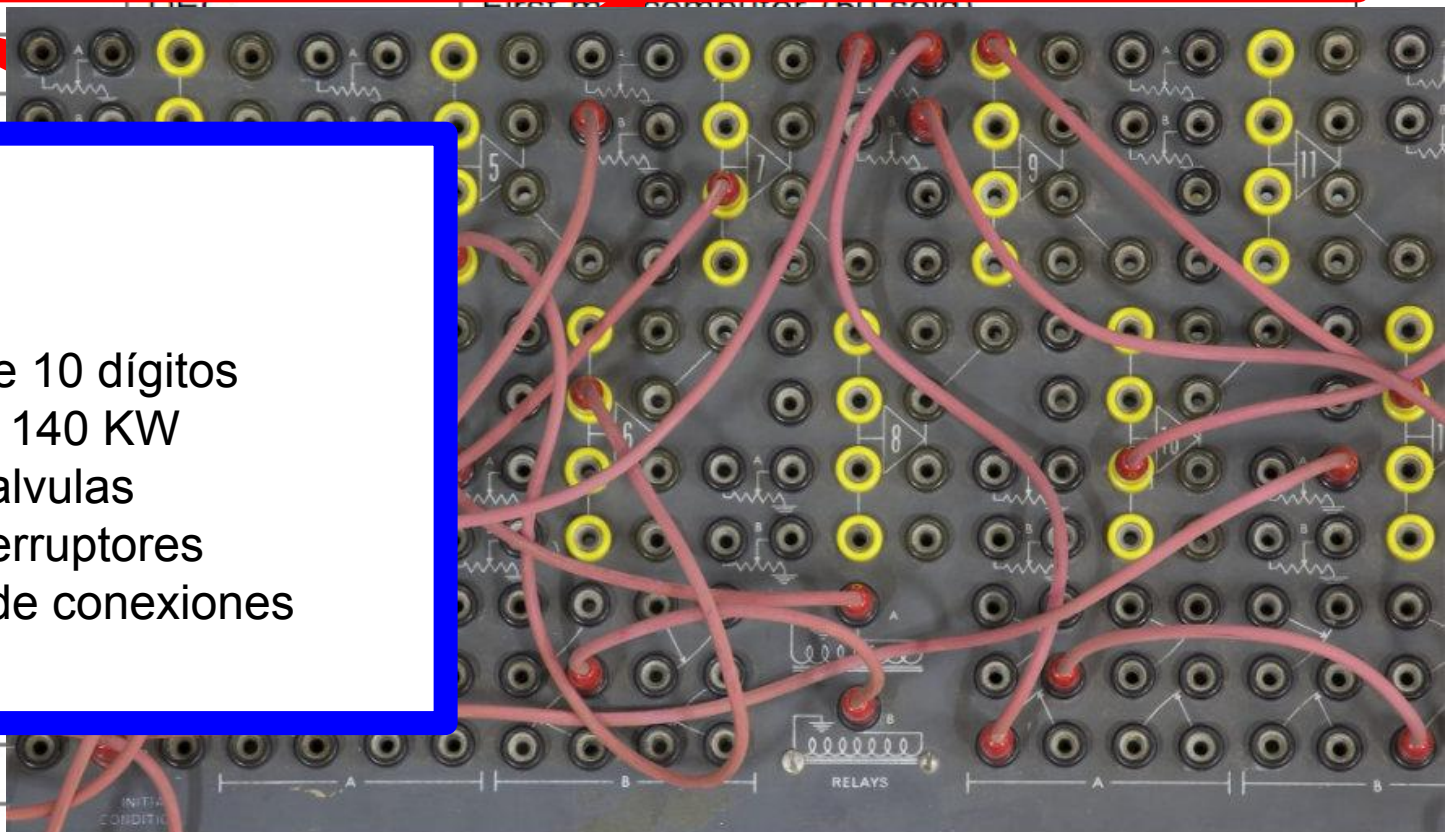
- 20 reg. de 10 dígitos
- 30 tons y 140 KW
- 18.000 valvulas
- 6.000 interruptores
- tableros de conexiones

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design

1960	PDP-1	DEC	First time computer has screen
1961	1401	IBM	First time computer has disk
1962	7090	IBM	First time computer has color screen

ENIAC

- 20 reg. de 10 dígitos
- 30 tons y 140 KW
- 18.000 valvulas
- 6.000 interruptores
- tableros de conexiones



1987	SPARC	Intel	First time computer has cache
1990	RS6000	IBM	First superscalar machine

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	First widely popular small business machine
1962	7094		First computer for scientific computing in the early 1960s
1963	B5000		First computer designed for a high-level language
1964	360		First computer as a family

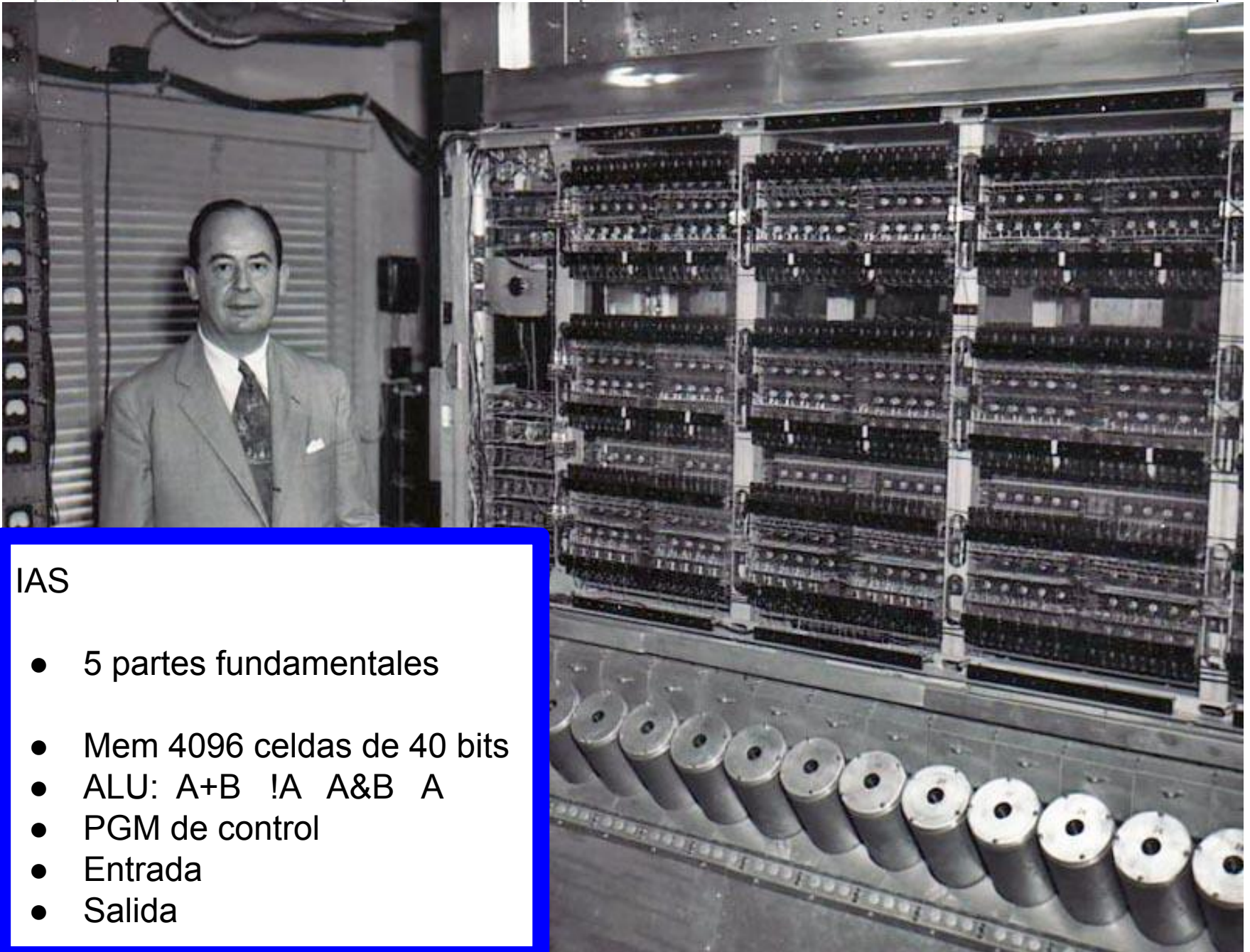
Dispositivos de Válvulas

IAS

- 5 partes fundamentales
- Mem 4096 celdas de 40 bits
- ALU: $A+B$ $A-B$ $A \times B$ A/B
- PGM de control
- Entrada
- Salida

- First mass-market minicomputer (50,000 sold)
- Dominated minicomputers in the 1970s
- First general-purpose 8-bit computer on a chip
- First vector supercomputer
- First 32-bit superminicomputer
- Started the modern personal computer era
- First commercial RISC machine
- First SPARC-based RISC workstation
- First superscalar machine

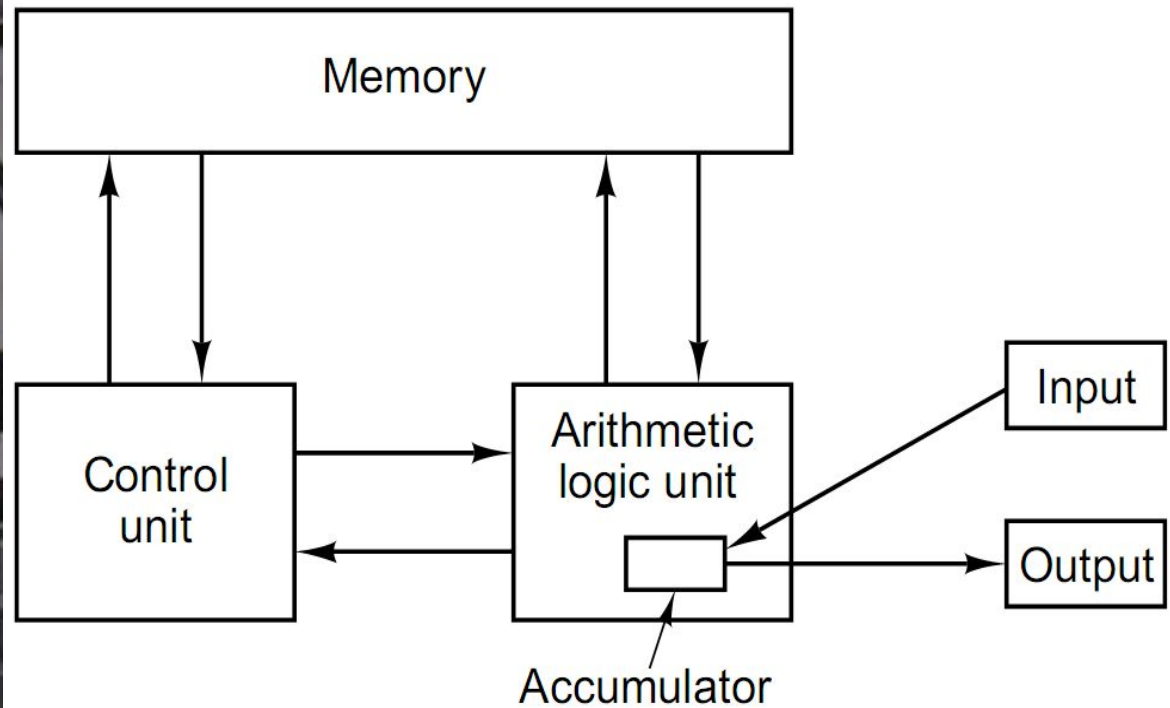
Year	Name	Made by	Comments
------	------	---------	----------



IAS

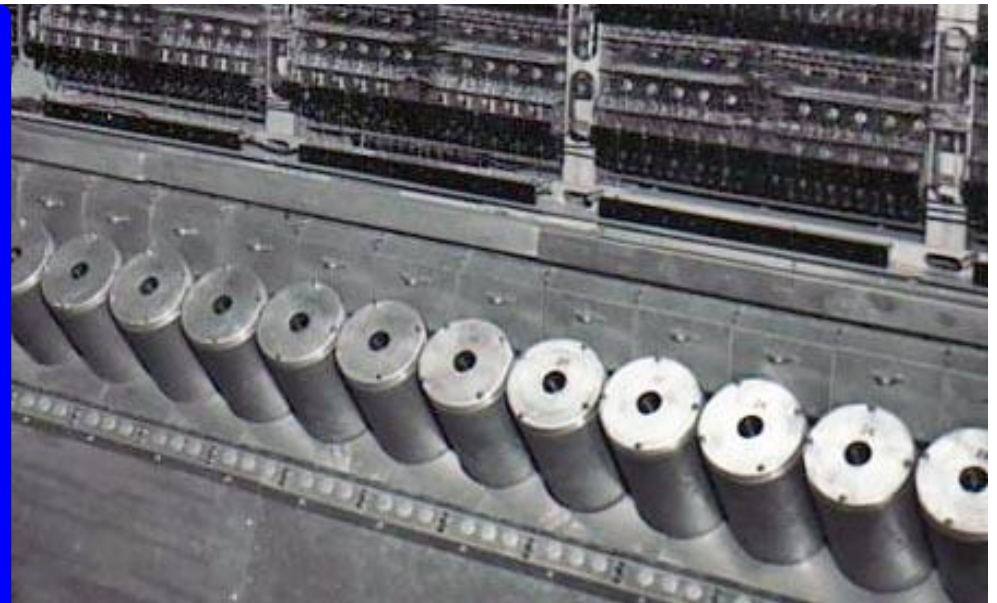
- 5 partes fundamentales
- Mem 4096 celdas de 40 bits
- ALU: $A+B$ $!A$ $A\&B$ A
- PGM de control
- Entrada
- Salida

Year	Name	Made by	Comments
------	------	---------	----------



IAS

- 5 partes fundamentales
- Mem 4096 celdas de 40 bits
- ALU: $A+B$ $!A$ $A\&B$ A
- PGM de control
- Entrada
- Salida



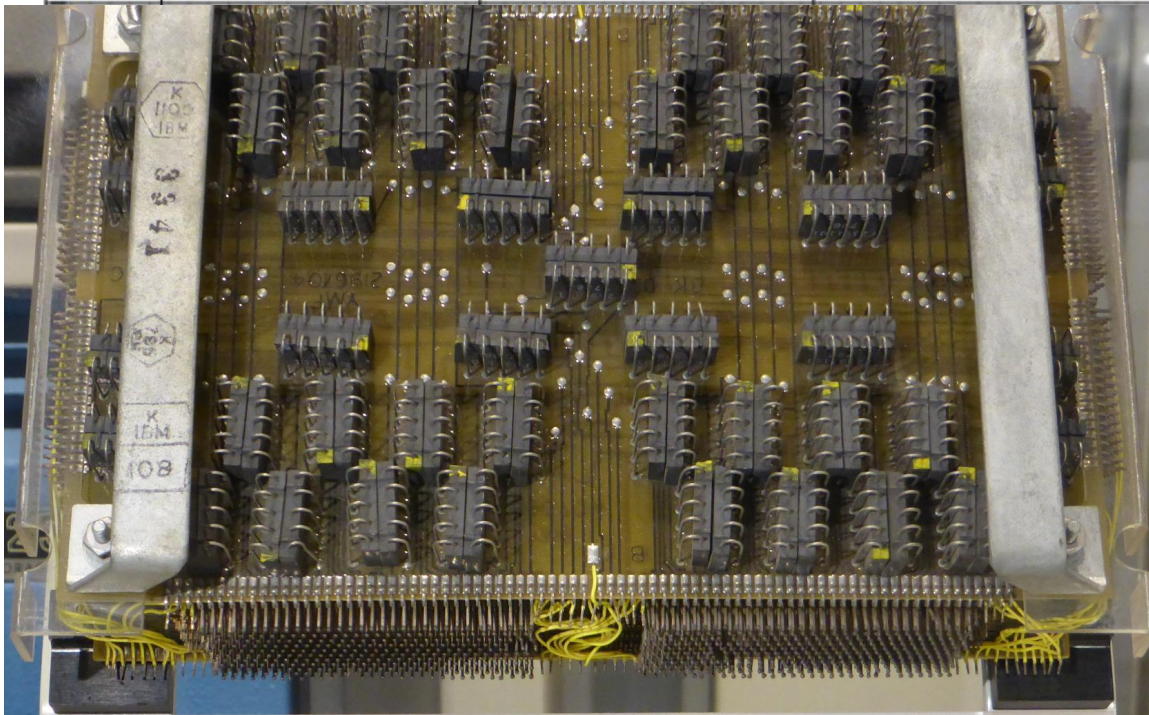
Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	Burroughs	First machine designed for a high-level language
1964	360	IBM	First 360-bit machine designed as a family
1964	6600	Control Data	First transistorized computer
1965	PDP-8	DEC	First 8-bit minicomputer (50,000 sold)
1970	PDP-11	DEC	Most popular minicomputer in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Dispositivos a
Transistores

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind		
1952	IAS		
1960	PDP-1		
1961	1401		
1962	7094		
1963	B5000		
1964	360		
1964	6600		
1965	PDP-8		
1970	PDP-11		
1974	8080		
1974	CRAY-		
1978	VAX		
1981	IBM PC		
1985	MIPS		
1987	SPARC		
1990	RS6000		



Year	Name	Made by	Comments
			ld a digital computer
			calculating machine
			puter
			eral-purpose computer
			history starts here
			m computer

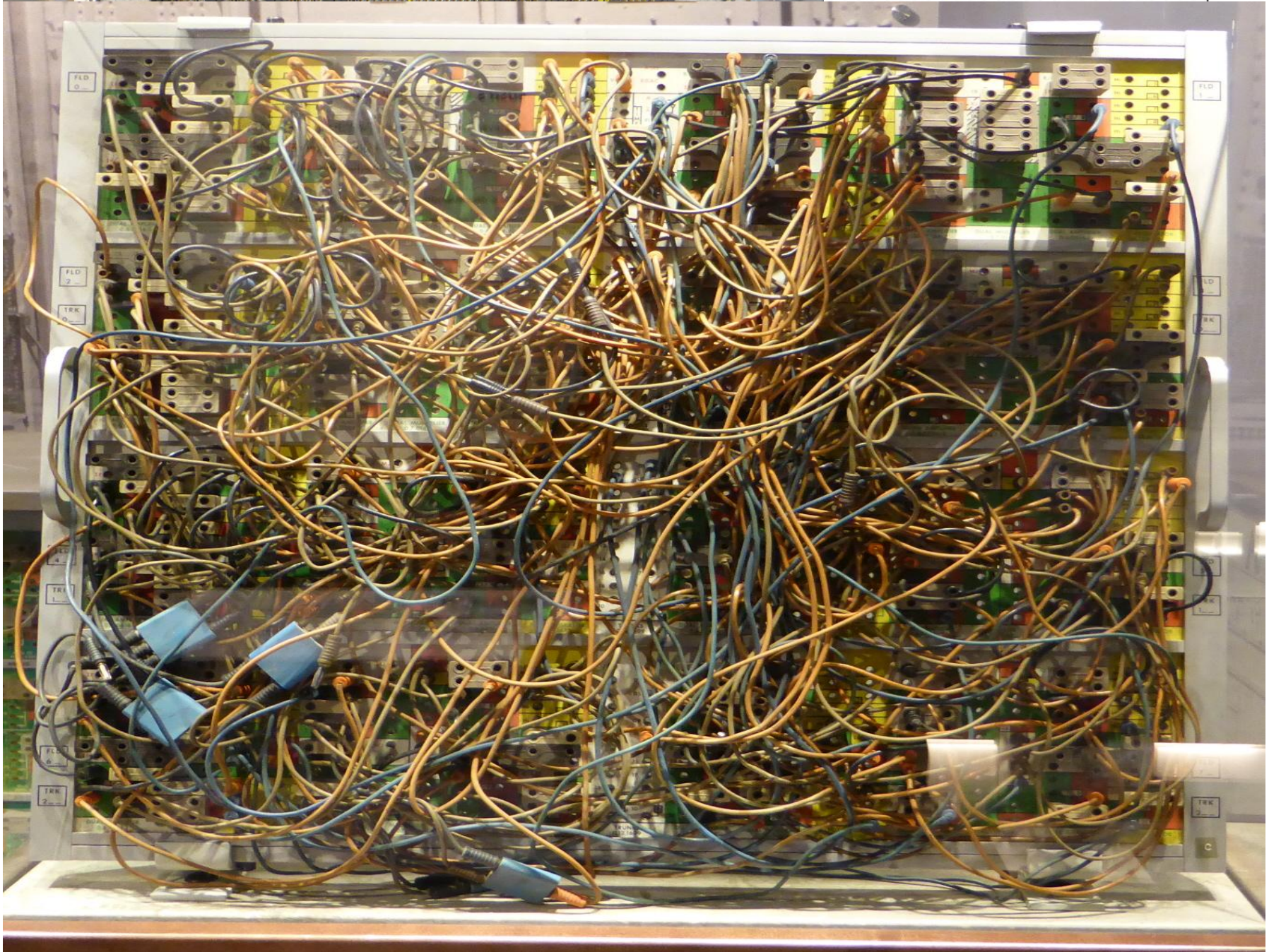


Magnetic Core Stack, IBM



1964	6000
1965	PDP-8
1970	PDP-11
1974	8080
1974	CRAY-
1978	VAX
1981	IBM PC
1985	MIPS
1987	SPARC
1990	RS6000

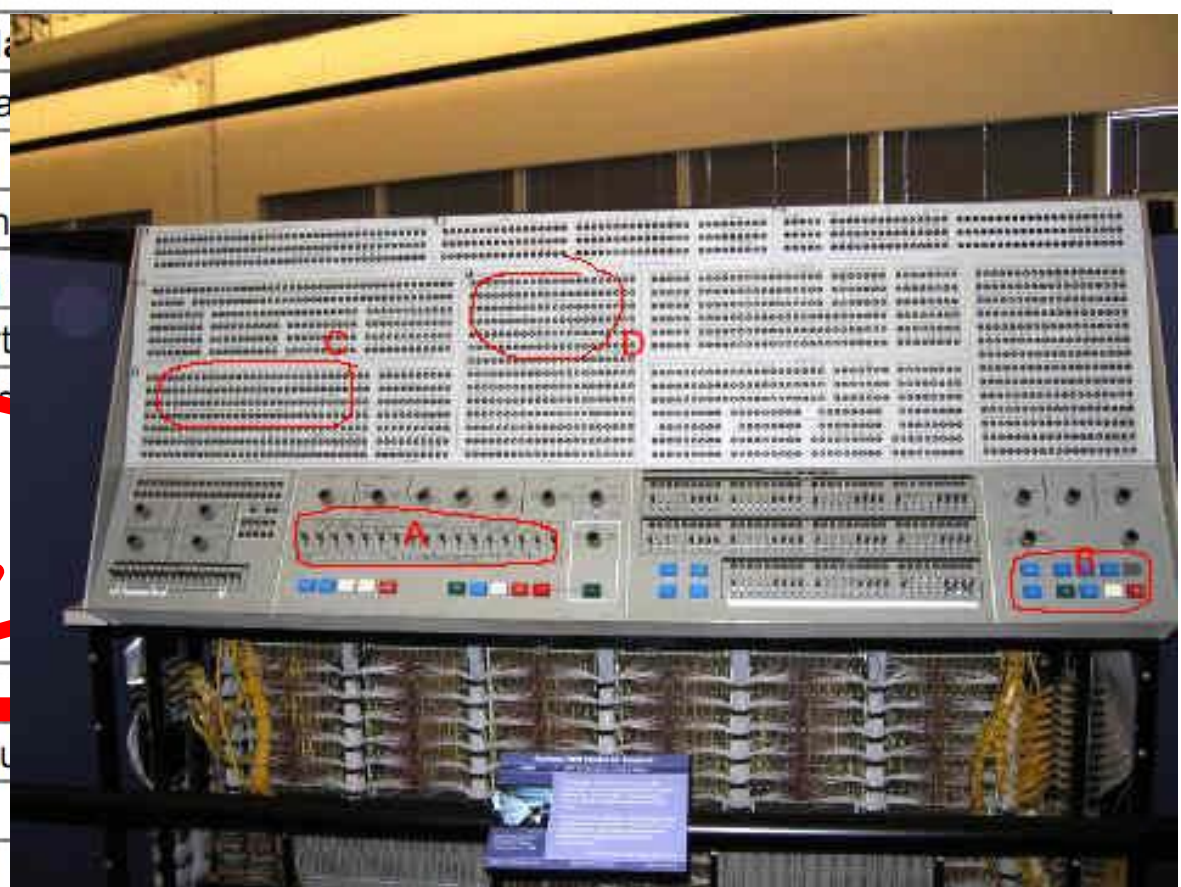
Year	Name	Made by	Comments
------	------	---------	----------



Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Cambridge	First stored-program computer
1951	Whirlwind I	MIT	First real-time computer
1952	IAS	IBM	First computer to use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Extremely popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	Burroughs	First machine designed for a high-level language
1964	360	IBM	First product line designed as a family
1964	6600	CDC	First scientific supercomputer
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Circuitos Integrados

Year	Name	Manufacturer
1834	Analytical Engine	Babbage
1936	Z1	Zuse
1943	COLOSSUS	British
1944	Mark I	Aiken
1946	ENIAC I	Eckert
1949	EDSAC	
1951	Whirlwind I	
1952	IAS	
1960	PDP-1	DEC
1961	1401	IBM
1962	7094	IBM



1963	B5000	Burroughs	
1964	360	IBM	
1964	6600	CDC	
1965	PDP-8	DEC	First mass-market minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

Year	Name	Ma
1834	Analytical Engine	Babba



Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	First American general-purpose computer
1946	ENIAC I	Eckert/Mauchley	Modern computer history starts here
1949	EDSAC	Wilkes	First stored-program computer
1951	Whirlwind I	M.I.T.	First real-time computer
1952	IAS	Von Neumann	Most current machines use this design
1960	PDP-1	DEC	First minicomputer (50 sold)
1961	1401	IBM	Enormously popular small business machine
1962	7094	IBM	Dominated scientific computing in the early 1960s
1963	B5000	IBM	Designed for a high-level language
1964	360	IBM	Designed as a family
1964	6600	IBM	Supercomputer
1965	PDP-8	DEC	First 8-bit minicomputer (50,000 sold)
1970	PDP-11	DEC	Dominated minicomputers in the 1970s
1974	8080	Intel	First general-purpose 8-bit computer on a chip
1974	CRAY-1	Cray	First vector supercomputer
1978	VAX	DEC	First 32-bit superminicomputer
1981	IBM PC	IBM	Started the modern personal computer era
1985	MIPS	MIPS	First commercial RISC machine
1987	SPARC	Sun	First SPARC-based RISC workstation
1990	RS6000	IBM	First superscalar machine

C.I. a gran escla (VLSI)

Year	Name	Made by	Comments
1834	Analytical Engine	Babbage	First attempt to build a digital computer
1936	Z1	Zuse	First working relay calculating machine
1943	COLOSSUS	British gov't	First electronic computer
1944	Mark I	Aiken	
1946	ENIAC I	Eckert	
1949	EDSAC	Wilkes	
1951	Whirlwind I	M.I.T.	
1952	IAS	Von Ne	
1960	PDP-1	DEC	
1961	1401	IBM	
1962	7094	IBM	
1963	B5000		
1964	360		

CRAY

- 15 GFlops
- Mem 2Gword
- 64 bits
- \$30.000.000

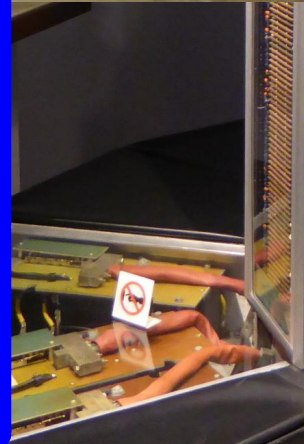


Year	Name	Made by	Comments
			ild a digital computer
			y calculating machine
			computer



CRAY

- 15 GFlops
- Mem 2Gword
- 64 bits
- \$30.000.000



Evolución

De la antigüedad a nuestros días

Arquitectura de computadoras

Pablo A. Montini
Juan I. Iturriaga
Franco Lanzillotta